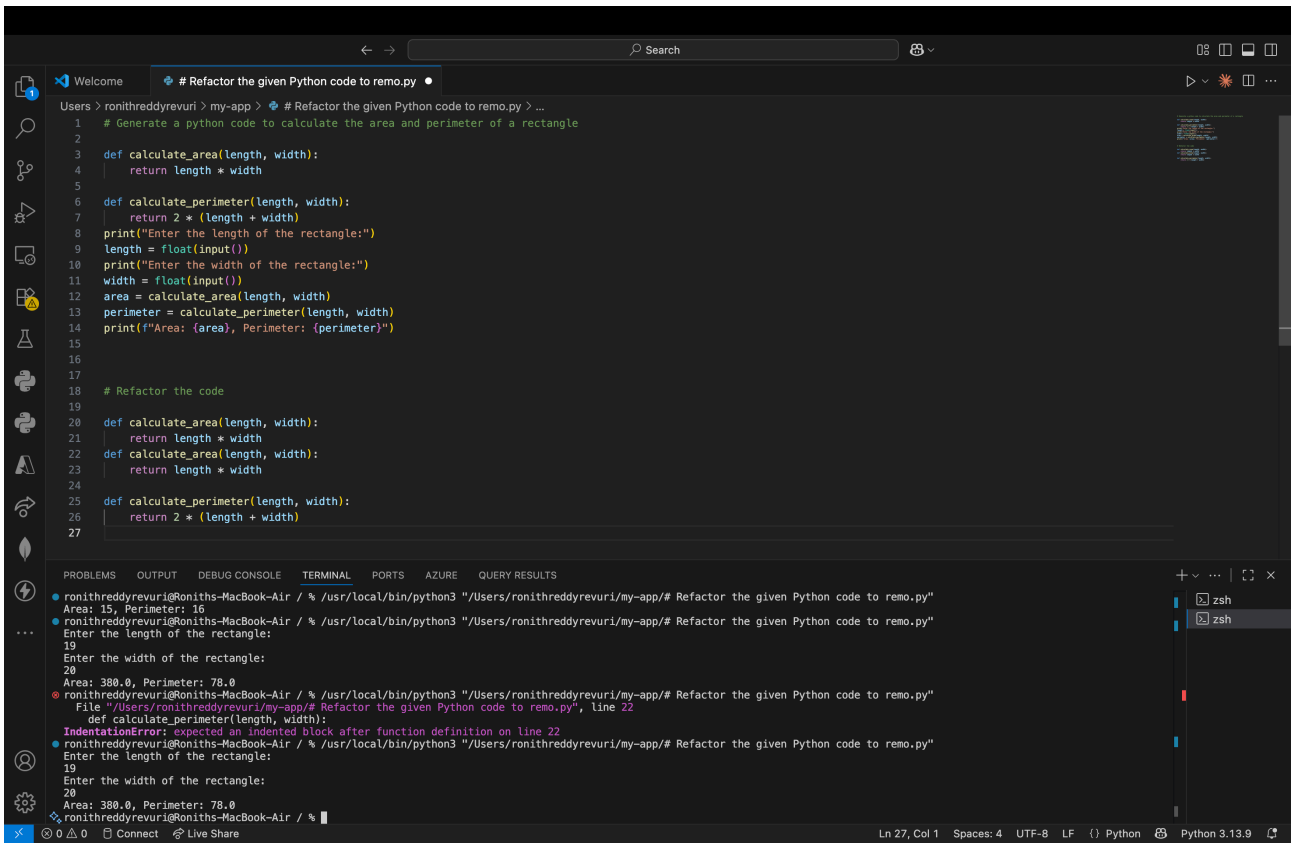
ASSIGNMENT- 13

TASK 1 – ELIMINATING REPETITIVE LOGIC

PROMPT :
“REFACTOR THE GIVEN PYTHON CODE TO REMOVE REPEATED CALCULATIONS AND CREATE A REUSABLE FUNCTION WITH PROPER DOCUMENTATION.”

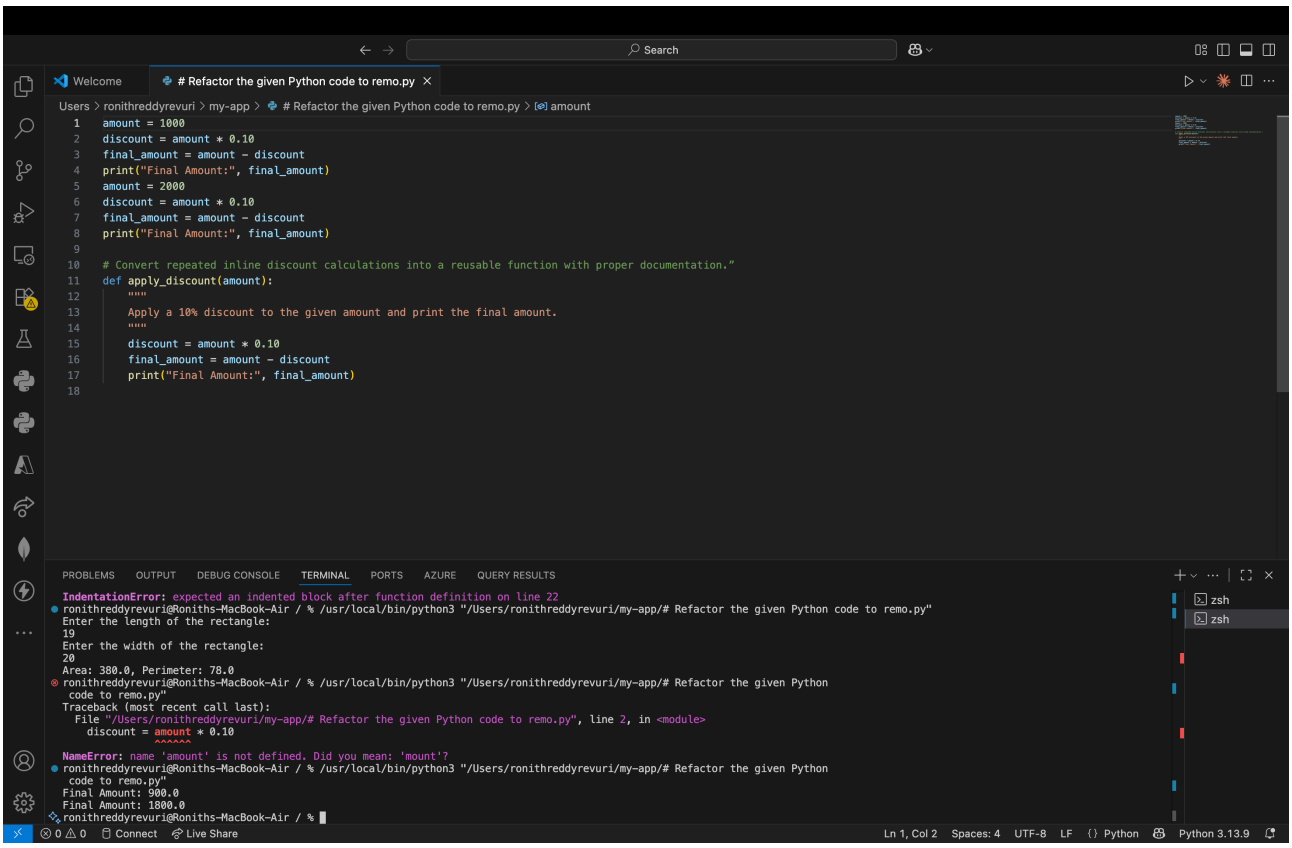
EXPLANATION :
THE ORIGINAL CODE DID THE CALCULATIONS OVER AND OVER. TO MAKE IT BETTER A NEW FUNCTION WAS MADE THAT CAN DO BOTH AREA AND PERIMETER CALCULATIONS. THIS NEW FUNCTION MAKES THE CODE LOOK NICER. IT IS ALSO EASIER TO USE THE CODE AGAIN WHEN IT IS NEEDED. THE CODE IS CLEANER NOW BECAUSE OF THIS FUNCTION THAT HANDLES AREA AND PERIMETER CALCULATIONS.



TASK 2 – MODULARIZING INLINE CALCULATIONS

PROMPT :
CONVERT REPEATED INLINE DISCOUNT CALCULATIONS INTO A REUSABLE FUNCTION WITH PROPER DOCUMENTATION.

EXPLANATION :
WE DID NOT WANT TO DO THE THING OVER AND OVER. SO WE MADE A FUNCTION TO WORK OUT THE AMOUNT. THIS MAKES THE CODE EASIER TO UNDERSTAND. IT ALSO HELPS WHEN WE NEED TO UPDATE THE RULES ON. THE FUNCTION REALLY HELPS WITH THIS. IT MAKES THE RULES EASIER TO UPDATE IN THE FUTURE. WE CAN JUST CHANGE THE FUNCTION. IT WILL WORK FOR THE FINAL AMOUNT.

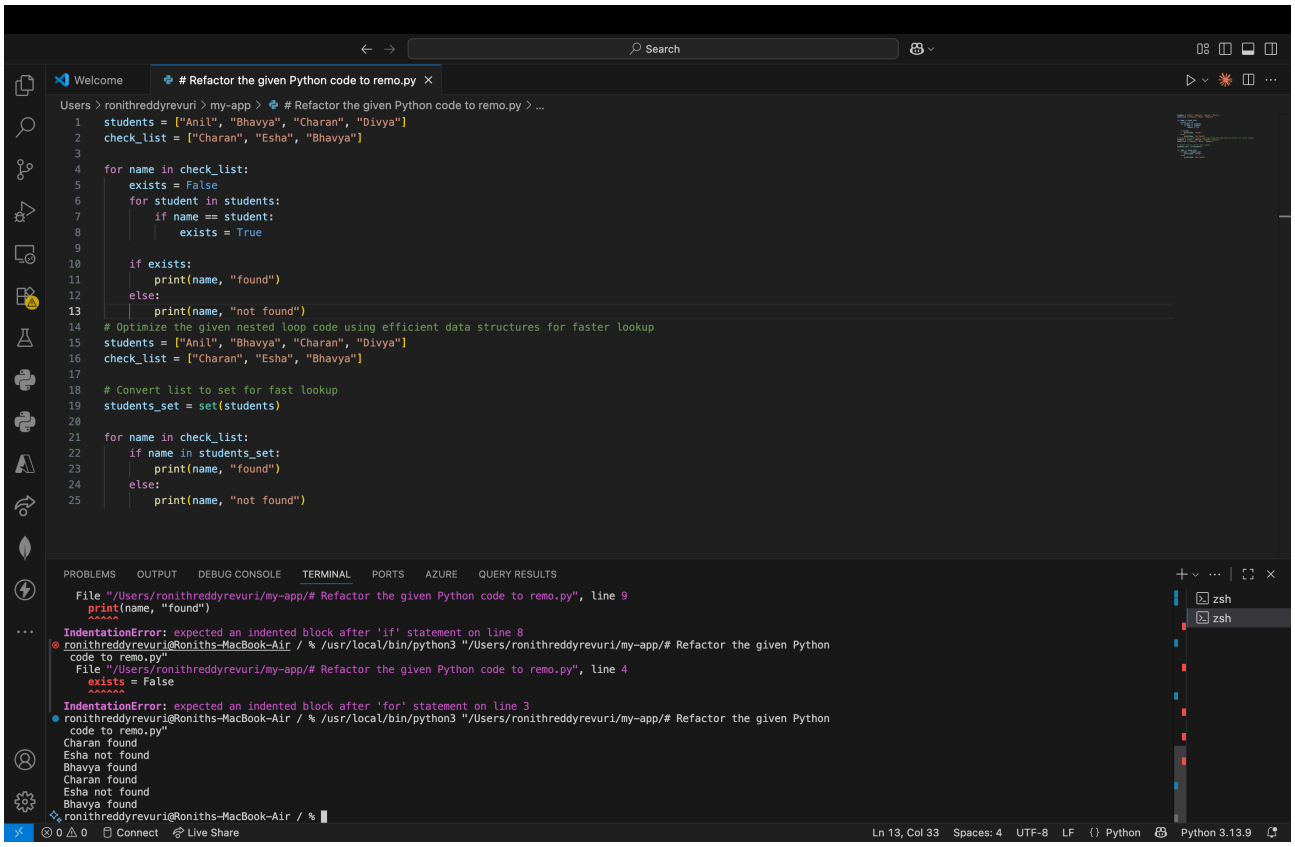


TASK 3 : IMPROVING LOOP PERFORMANCE

PROMPT :
OPTIMIZE THE GIVEN NESTED LOOP CODE USING EFFICIENT DATA STRUCTURES FOR FASTER LOOKUP

EXPLANATION :

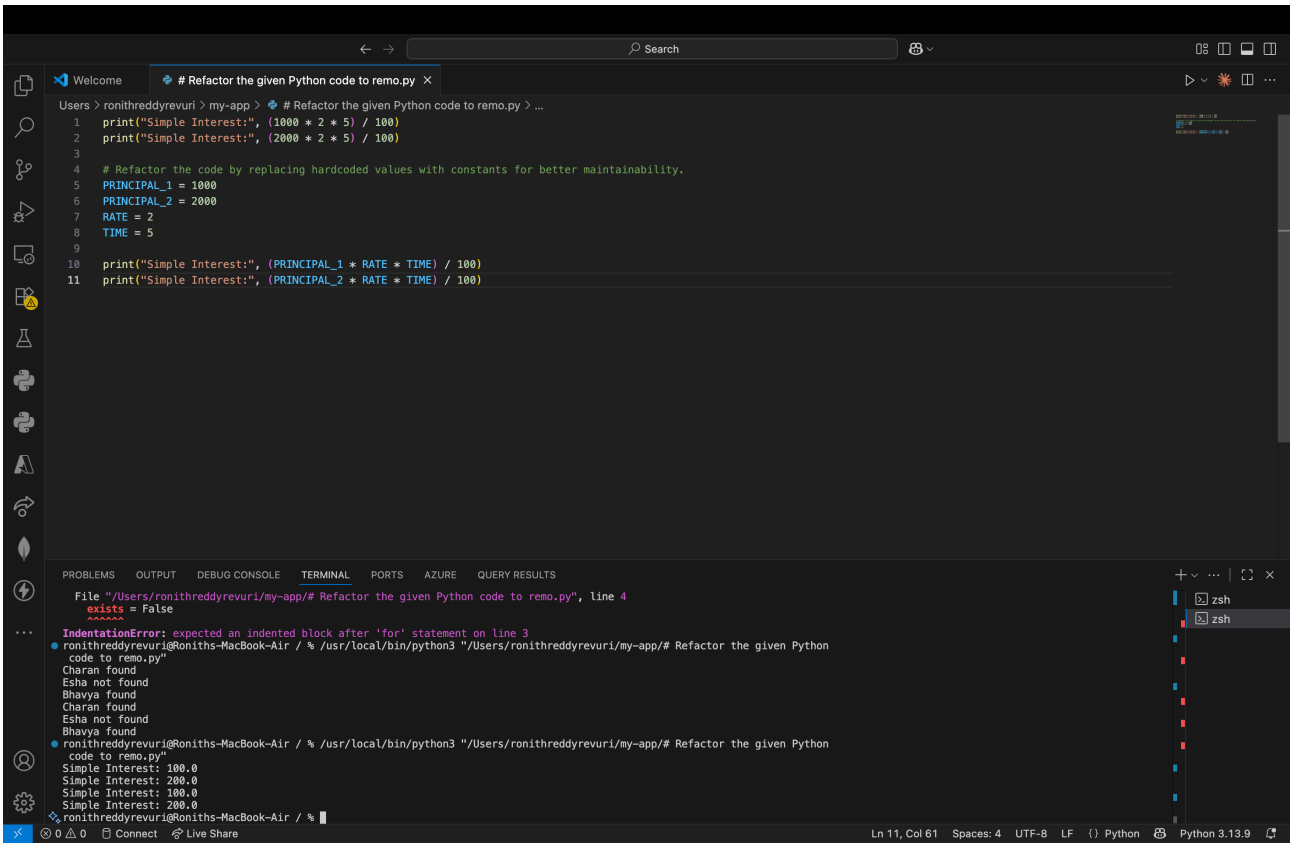
THE ORIGINAL CODE WAS USING LOOPS WHICH ARE REALLY SLOW. WE CHANGED IT TO USE A SET FOR SEARCHING BECAUSE THAT IS MUCH FASTER. THIS MAKES THE TIME IT TAKES TO DO THINGS A LOT AND THE CODE WORKS BETTER. WE USED A SET FOR SEARCHING SO THE CODE CAN FIND THINGS QUICKLY.



TASK 4 – REPLACING HARDCODED VALUES

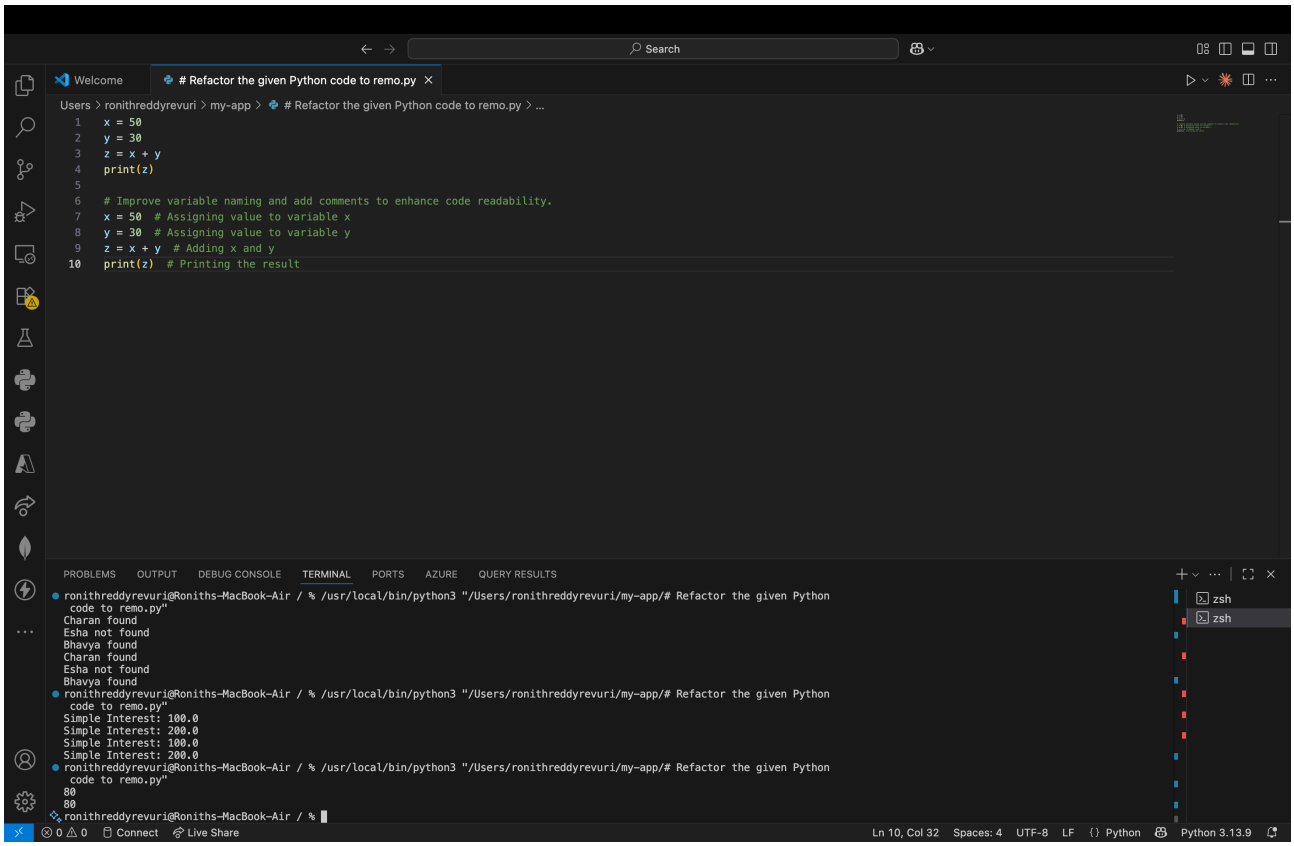
PROMPT :
REFACTOR THE CODE BY REPLACING HARDCODED VALUES WITH CONSTANTS FOR BETTER MAINTAINABILITY.

EXPLANATION :
WE REPLACED FIXED NUMBERS WITH CONSTANTS, LIKE RATE AND TIME. THIS MAKES THE CODE SIMPLER TO CHANGE AND UNDERSTAND.IT ALSO HELPS WITH UPKEEP.USING THESE CONSTANTS MAKES IT EASIER TO UPDATE THE CODE.



TASK 5 : IMPROVING VARIABLE NAMING

PROMPT :
IMPROVE VARIABLE NAMING AND ADD COMMENTS TO ENHANCE CODE READABILITY.

EXPLANATION :

THE ORIGINAL VARIABLES WERE NOT CLEAR. THEY WERE CALLED X, Y Z. WE CHANGED THEM TO NAMES THAT MAKE SENSE. THIS HELPS WITH UNDERSTANDING THE CODE. WE ALSO ADDED COMMENTS TO EXPLAIN WHAT THE LOGIC IS. THE VARIABLES NOW HAVE NAMES THAT'RE EASY TO UNDERSTAND.